OPERATING SYSTEMS

UE25CS242A

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File Management

Pavan A C

Department of Computer Science & Engineering

OPERATING SYSTEMS

Mass-Storage Structure

Pavan A C
Department of Computer Science & Engineering

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Slides Credits for all PPTs of this course



- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:

1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
3. Some presentation transcripts from A. Frank – P. Weisberg
4. Some conceptual text from Operating Systems: Three Easy Pieces, Remzi Arpaci-Dusseau, Andrea Arpaci Dusseau

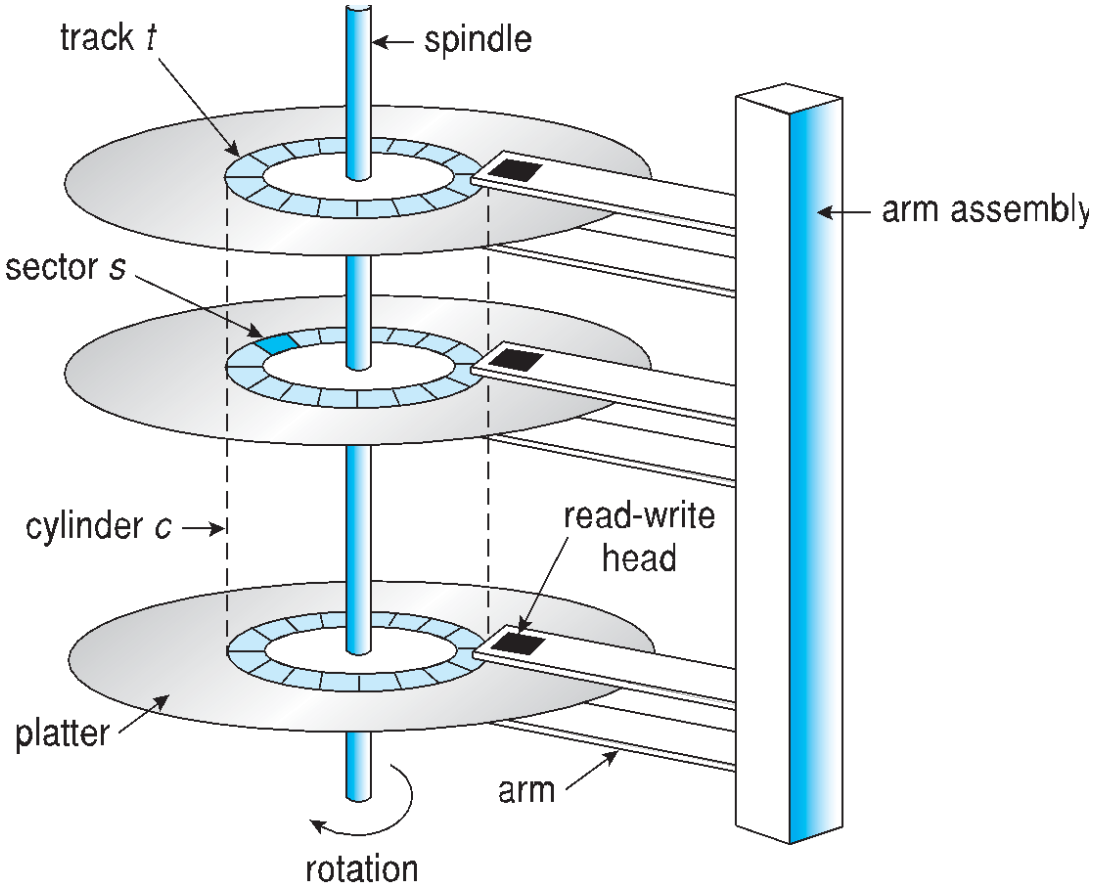
Overview of Mass Storage Structure

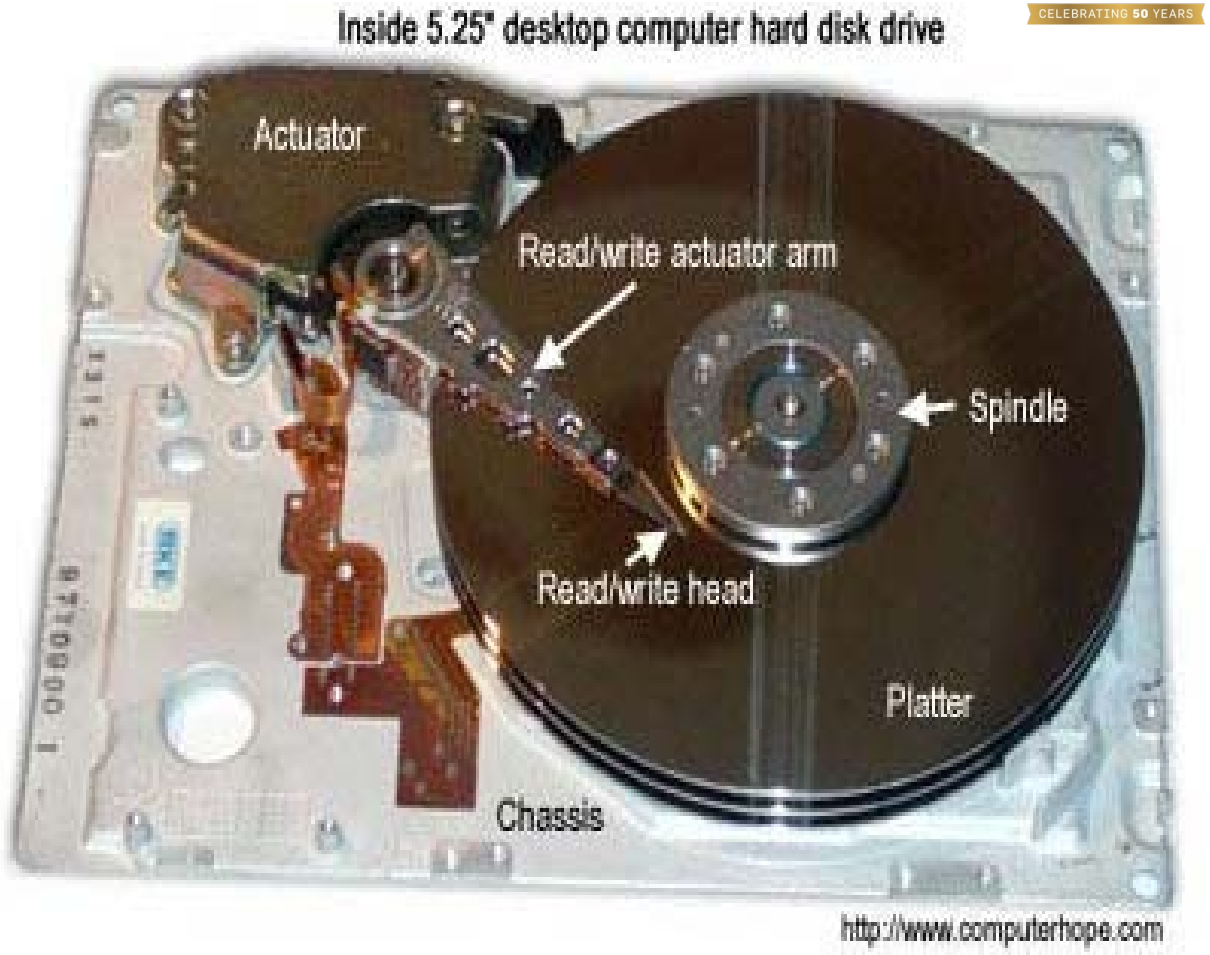
- **Magnetic disks** provide bulk of secondary storage of modern computers.
- Each disk platter has a flat circular shape, like a CD. Common platter diameters range from 1.8 to 3.5 inches.
- The two surfaces of a platter are covered with a magnetic material.
- The heads are attached to a disk arm that moves all the heads as a unit.
- The surface of a platter is logically divided into circular tracks, which are subdivided into sectors.
- The set of tracks that are at one arm position makes up a cylinder.

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Moving-head Disk Mechanism





Overview of Mass Storage Structure

- When the disk is in use, a drive motor spins it at high speed. Most drives rotate 60 to 250 times per second.

Disk speed has two parts.

- The transfer rate is the rate at which data flow between the drive and the computer.
- The positioning time, or random-access time, consists of two parts
  - The time necessary to move the disk arm to the desired cylinder, called the **seek time**.
  - the time necessary for the desired sector to rotate to the disk head, called the **rotational latency**.

Disk Latency = Seek time + Rotational latency + Transfer time

Overview of Mass Storage Structure



- Disk head flies on an extremely thin cushion of air.
- Head will sometimes damage the magnetic surface when it makes contact with it.
  - Head crash
  - Not recoverable
- A disk can be **removable.**
- Removable magnetic disks generally consist of one platte
- Examples CDs, DVDs, and Blu-ray discs as well as removable flash-memory

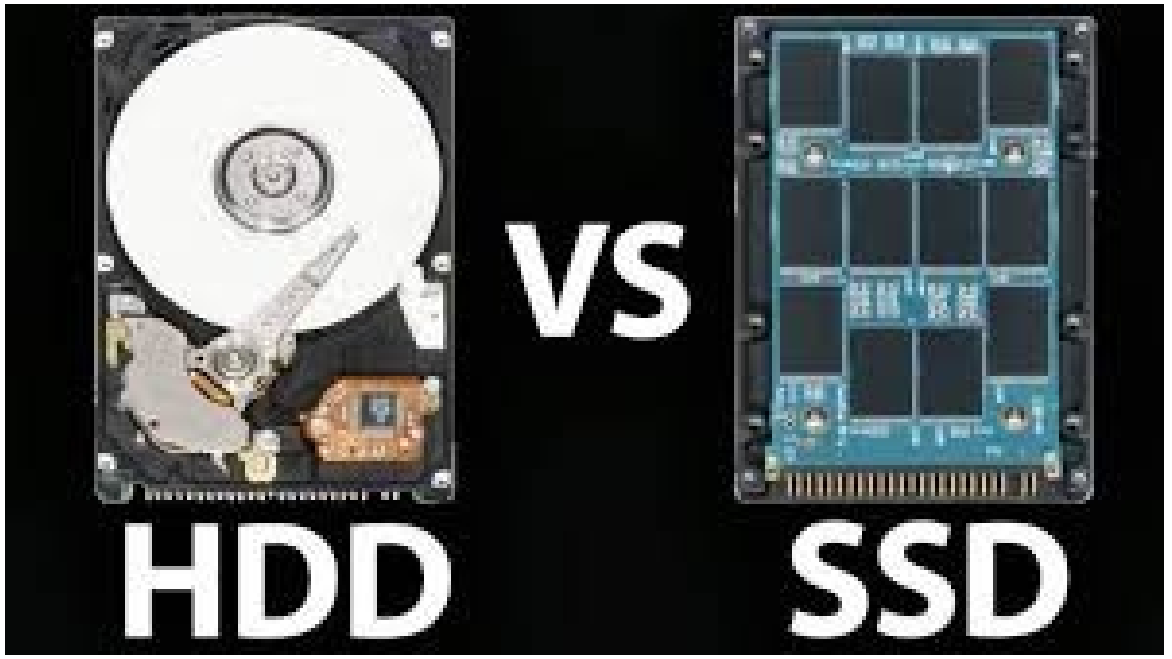
Mass Storage Structure (Cont.)

- Drive attached to computer via **I/O bus**
  - Busses vary, including **EIDE**, **ATA**, **SATA**, **USB**, **Fiber Channel**, **SCSI**, **SAS**, **Firewire**
  - **Host controller** in computer uses bus to talk to **disk controller** built into drive or storage array

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Solid-State Disks

- Nonvolatile memory used like a hard drive
- Can be more reliable than HDDs
- More expensive per MB
- Maybe have shorter life span
- Less capacity
- But much faster
- Standard Bus interfaces can be too slow -> the system bus (PCI, for example)
- No moving parts, so no seek time or rotational latency



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Magnetic Tape

- Was early secondary-storage medium
- Relatively permanent and holds large quantities of data
- Access time slow
- Random access ~1000 times slower than disk
- Mainly used for backup, storage of infrequently-used data, transfer medium between systems



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THANK YOU

Pavan A C

Department of Computer Science and Engineering

pavanac@pes.edu